



Reference-Free Assessment of Physical Consistency in World Model-based Video Generation

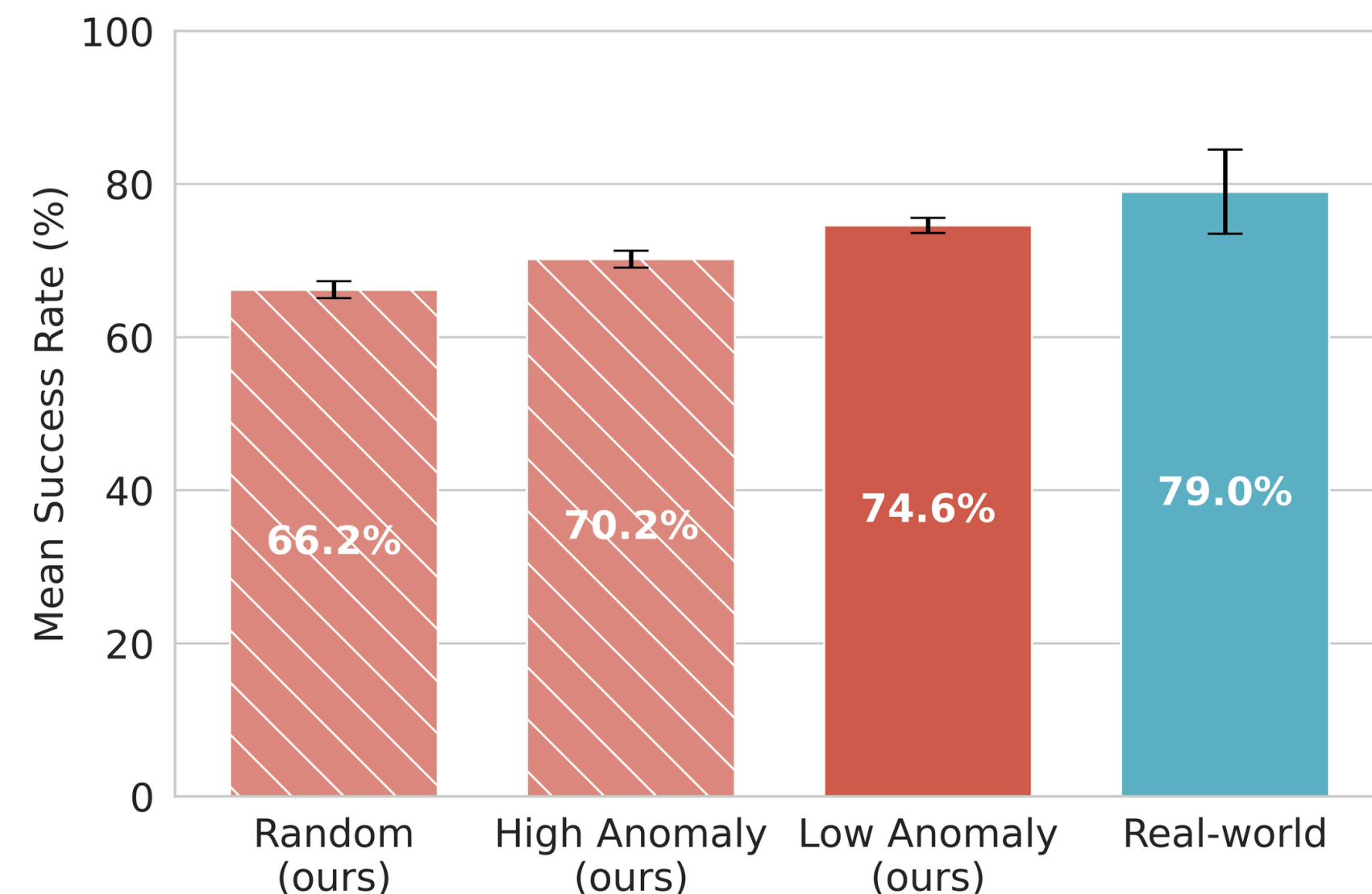
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Summary

Problem. VLA Testing with generated videos is viable, but falls short due to **artifacts**.

Question. Can we filter unreliable generated videos **without reference** so that test results are more aligned with the real world assessment?

Motivation. Physically reliable videos have different consistency scores from unreliable ones.



Key Idea

How can we calculate a video's physical consistency?

- By **comparing peers** → **Relative Assessment**
- By **comparing adjacent frames and pixels** → **Absolute Assessment**

Methods

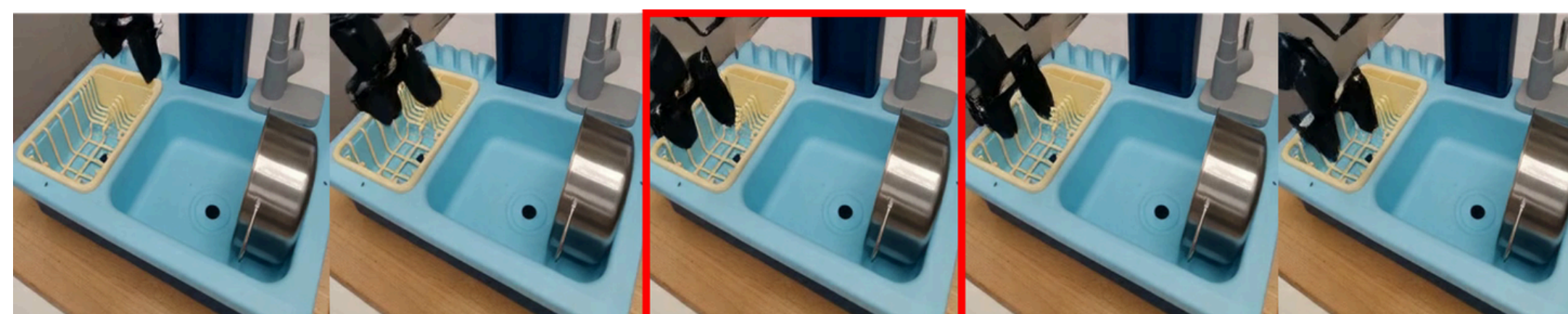
1. Relative Assessment

- generate multiple videos for the same situation
- evaluate each video
 - **3d consistency** with DROID-SLAM
 - **Photometric consistency** with SEA-RAFT
- **weight the two consistencies based on Shannon Entropy**

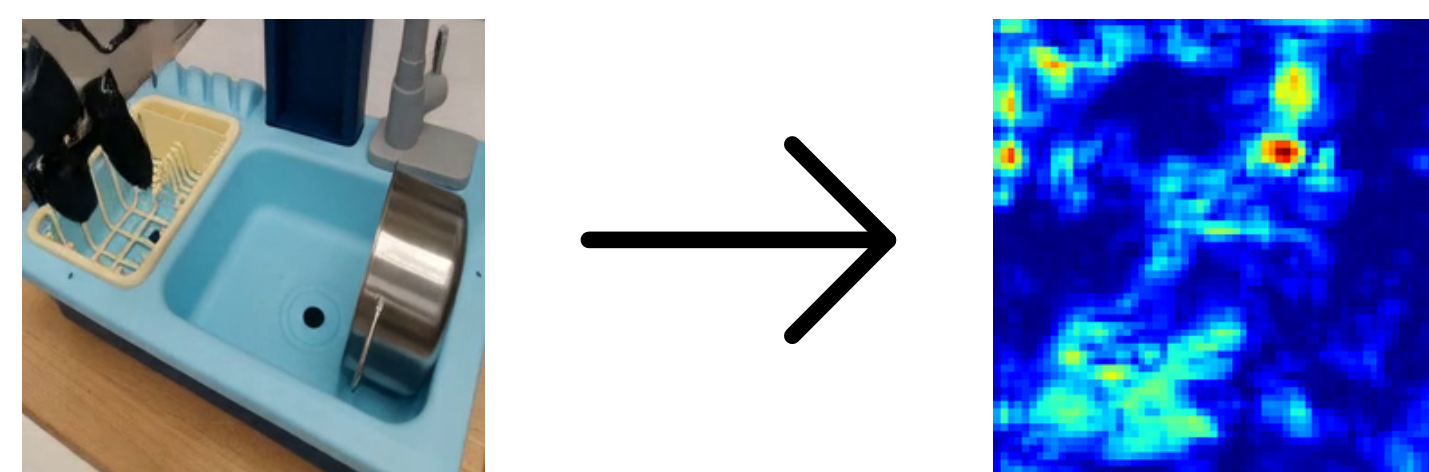
2. Absolute Assessment

- **Temporal Detection** finds the frame where **reprojection error from the previous frame is largest**

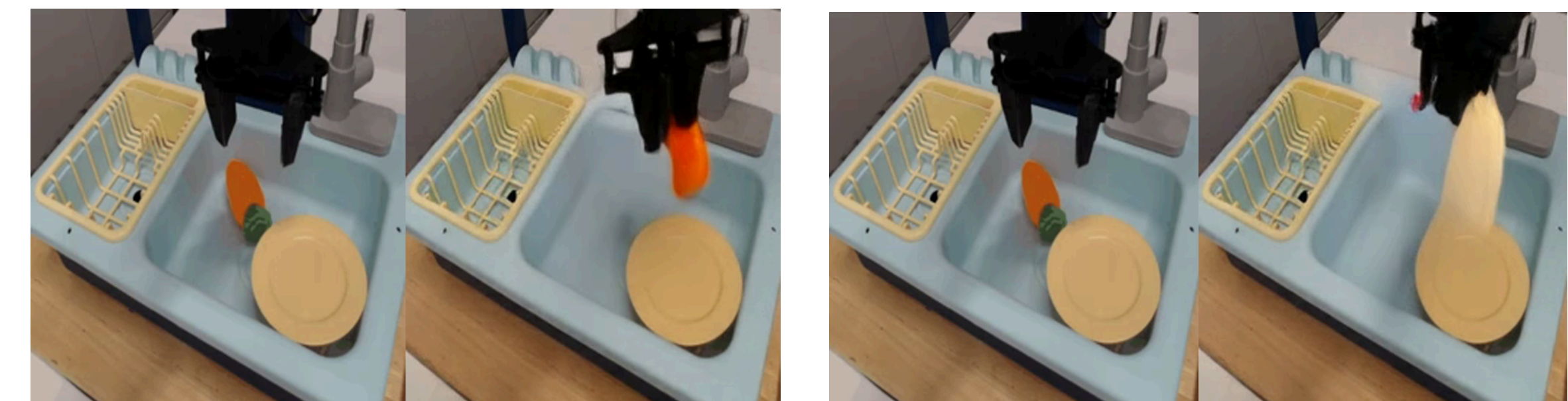
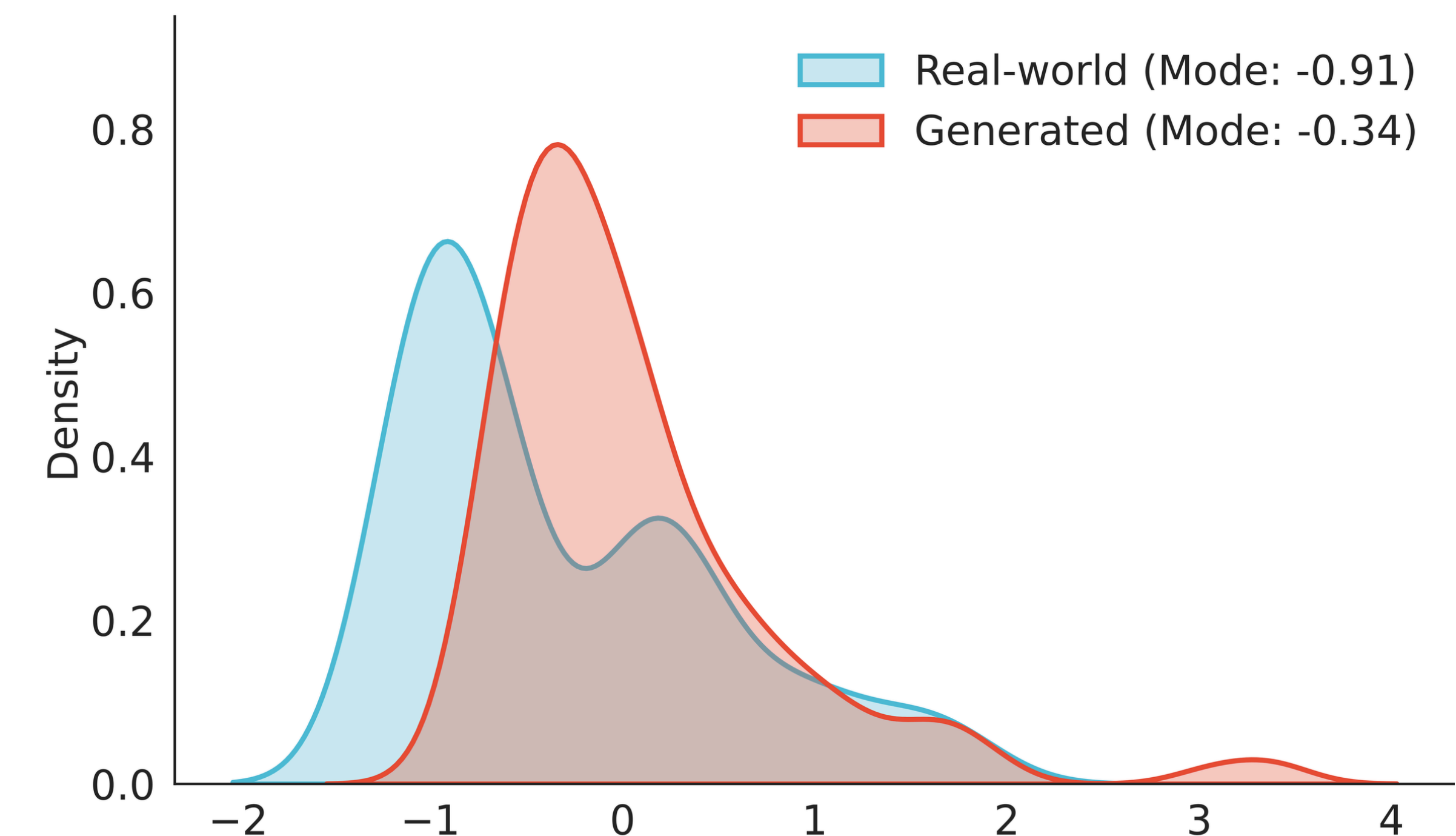
$$j_t = |e_t - e_{t-1}|, \quad t^* = \arg \max_t (j_t)$$



- **Spatial Localization** examines the **residual flow** for each edge **on the factor graph in DROID-SLAM** and calculates the pixel-wise magnitude as L2 norm to **visualize error values**



Results



Our finding.

- Generated videos with lower relative score (less artifacts) are more similar to real-world videos.
- **Physical artifacts in generated videos can be filtered by consistency scores alone, without human inspection or ground truth.**